Newton's *Principia* – the book that created mechanics

Philosophiæ Naturalis Principia Mathematica – "The Mathematical Principles of Natural Philosophy" – appeared in London in the summer of 1687. In some 500 Latin pages Isaac Newton showed that the same few laws govern an apple, a tide, the Moon and a comet. Here the book is summarised part by part, and you can try out the three laws of motion and the law of areas yourself in four interactive 3D animations.

How the book came about

In August 1684 the young astronomer Edmond Halley travelled to Cambridge to put a question to Newton that he, Christopher Wren and Robert Hooke had been unable to answer: what path does a planet follow if the Sun pulls on it with a force that falls off as the square of the distance? Newton answered at once: an ellipse – he had calculated it. He could not find his papers, but promised to send the proof. In November Halley received a short nine-page manuscript, *De motu corporum in gyrum* ("On the motion of bodies in orbit"). Halley realised what he had in his hands and persuaded Newton to write the whole thing out. It took a good eighteen months and grew into three books.

1684	Halley's visit; *De motu* is sent to the Royal Society.
1686	Book I is presented to the Royal Society. Its president, Samuel Pepys, gives his *imprimatur* on 5 July. The Society has no money (it has been spent on a book on the history of fishes), so Halley pays for the printing out of his own pocket. Hooke demands credit for the inverse-square law, and Newton threatens to suppress Book III.
1687	The book appears at the beginning of July, in a print run of 250–400 copies.
1713	Second edition, edited by Roger Cotes. The famous *Scholium generale* is added.
1726	Third edition, edited by Henry Pemberton – the year before Newton's death.
1729	Andrew Motte's English translation. Émilie du Châtelet's French translation with commentary appears in 1756.

The book is written like Euclid's *Elements*: definitions, axioms (the three laws) and then propositions with proofs. Almost all the proofs are geometrical. Newton had invented his method of fluxions (the differential calculus) decades earlier, but he writes in the classical geometrical limits – "first and last ratios" of vanishing quantities – so that nobody could reject the results because of a new and controversial method.

The definitions and the scholium on space and time

The book opens with eight definitions. The most important are:

Def. I–II: mass and momentum

The quantity of matter (*quantitas materiae*) is the density times the volume – what we today call the **mass** *m*. It is known by the weight, and Newton has shown with pendulum experiments that weight is proportional to mass. The quantity of motion (*quantitas motus*) is the velocity times the quantity of matter: $\boldsymbol{p = m \cdot v}$. When Newton speaks of "motion" in the laws, this is the quantity he means.

Def. III–V: forces

The "innate force" of matter (*vis insita*) is its resistance to a change of state – what we call **inertia**. An "impressed force" (*vis impressa*) is an action that changes a body's state; it lasts only as long as the action lasts. A **centripetal force** (Newton's own coinage) draws a body towards a centre – like gravity, magnetism or the cord of a sling.

Next comes a scholium (a commentary) on **absolute time, space and motion**. Absolute time "flows equably" without relation to anything external; absolute space remains always similar and immovable. We can only measure relative motions – but Newton holds that absolute rotation can be detected. His proof is the **bucket experiment**: hang a bucket of water from a twisted rope and let it go. At first the bucket turns while the water stays still, and the surface is flat. Then the water catches up with the bucket and the surface becomes concave. The relative motion between water and bucket is greatest at the start, yet the curvature appears only when the water itself rotates. So it is rotation relative to *space* that counts. Ernst Mach attacked the argument in 1883 (perhaps it is rotation relative to the fixed stars), and the debate helped inspire Einstein.

Axiomata sive leges motus – the three laws

After the definitions come the "Axioms, or Laws of Motion". They are the foundation of the book: everything that follows is proved from them. The first law goes back essentially to Galileo and Descartes, but brought together, precisely stated and used as the basis for the whole of nature, they are Newton's.

LEX I – THE LAW OF INERTIA

Corpus omne perseverare in statu suo quiescendi vel movendi uniformiter in directum, nisi quatenus a viribus impressis cogitur statum illum mutare.

Every body perseveres in its state of rest, or of uniform motion in a right line, unless it is compelled to change that state by forces impressed thereon.

Newton's own examples are projectiles, which are slowed only by air resistance and pulled down by gravity, a spinning top, which stops only because of the air, and planets and comets, which keep their motion for a long time in almost empty space. The difficulty with the law is that we never see it in

everyday life: everything we push comes to a halt. Aristotle therefore concluded that motion needs a force. The animation shows why he was wrong – and why it is so easy to believe it.

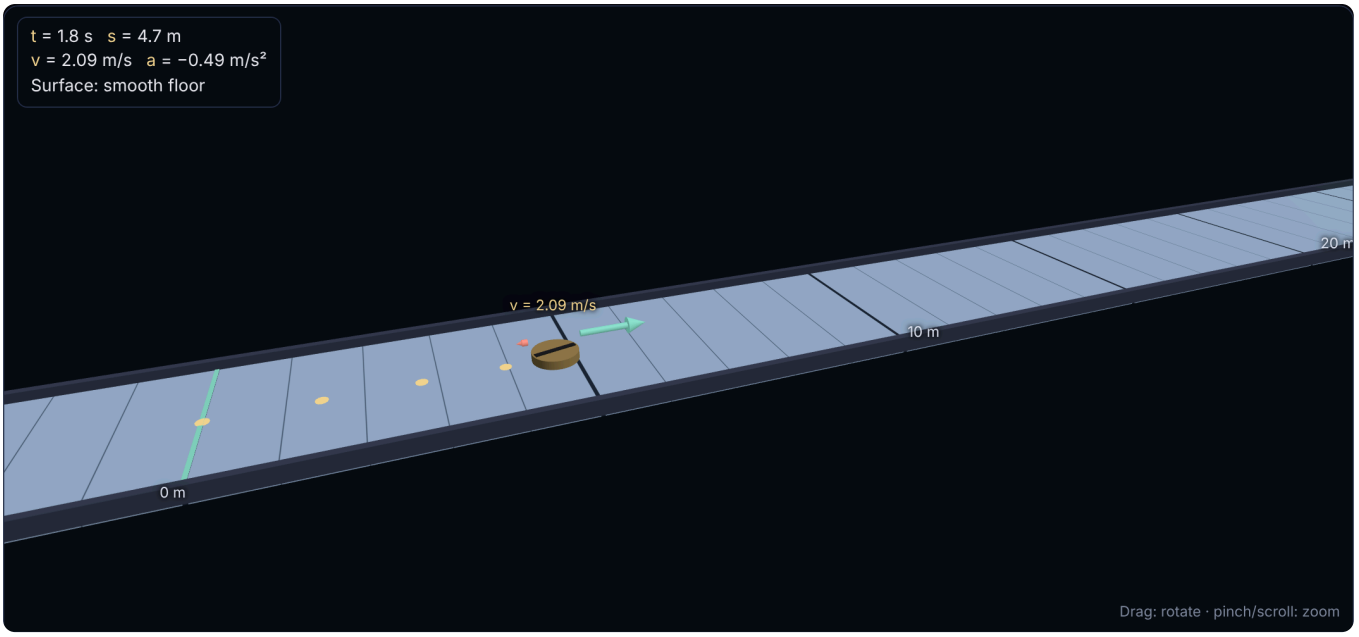


Figure 1. Lex I – the law of inertia – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/principia-en.html#lex1

Try it yourself. The marks on the track are placed every half second.

1. Set μ = 0. Why are the marks equally spaced all the way? Why does the puck never stop?
2. Turn μ up. What happens to the spacing of the marks? Which force is changing the puck's state?
3. Run it in reverse with friction: the puck accelerates from rest. Why does that look wrong – and what does it tell us about friction and the direction of time?
4. The lorry: no force pushes the crate forwards. Why does it look that way to the driver? (See the gold and blue marks at the roadside.)

LEX II – THE LAW OF FORCE

Mutationem motus proportionalem esse vi motrici impressae, et fieri secundum lineam rectam qua vis illa imprimitur.

The change of motion is proportional to the motive force impressed, and is made in the direction of the right line in which that force is impressed.

Notice that it does not say $F = m \cdot a$. Newton thinks in **impulses**: a force produces a change of momentum, and whether it acts "altogether and at once, or gradually and successively", the total result is the same. In modern language Newton's law is $\boldsymbol{\Delta p = F \cdot \Delta t}$ (the impulse theorem), and only in the limit of infinitely small, infinitely frequent impulses does it become $F = \mathrm{d}p/\mathrm{d}t = m \cdot a$. Leonhard Euler wrote that form down explicitly in 1750. In the animation you can switch between a constant force and Newton's impulses every half second – and watch the staircase in the graph approach the straight line.

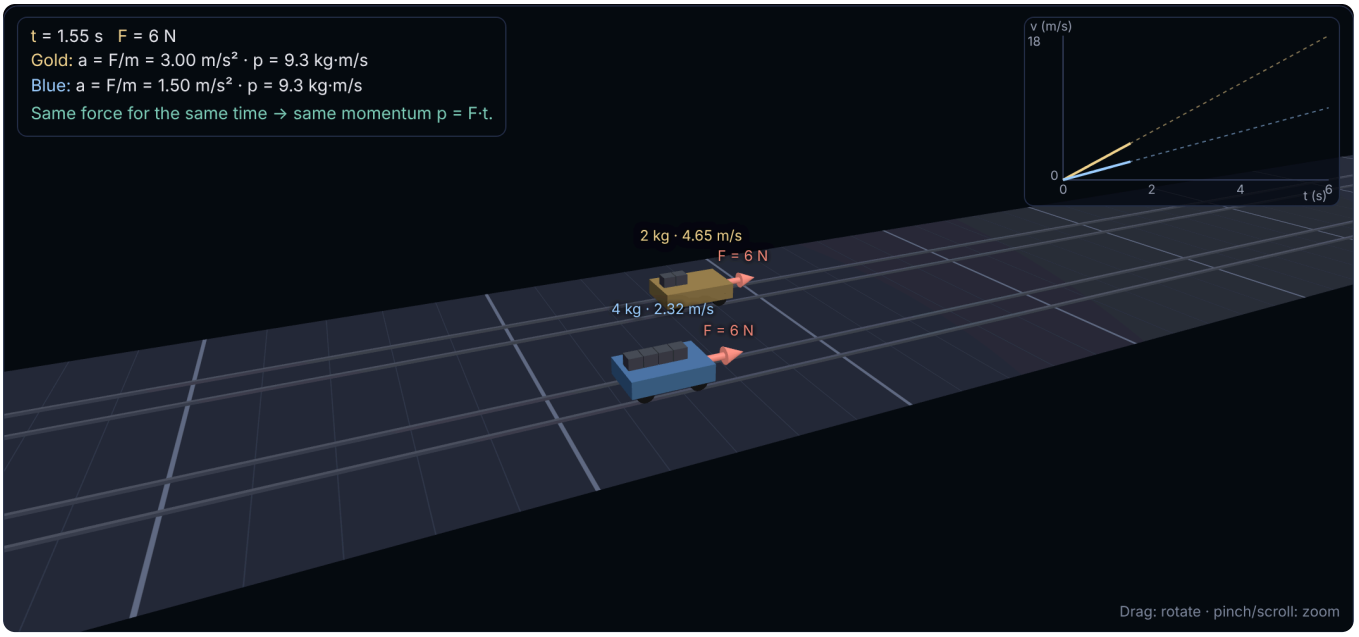


Figure 2. Lex II – the law of force – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/principia-en.html#lex2

Try it yourself. The two trolleys are pulled by equal forces on a frictionless track.

1. Give the blue trolley twice the mass of the gold one. How do the slopes in the *v*–*t* graph compare?
2. Read off the momentum $p = m \cdot v$ of both trolleys. Why is it the same, even though the speeds differ?
3. Switch to impulses. Why does the staircase always lie above the dashed line, yet meet it at the end of each interval? What happens if the impulses come closer together?

LEX III – THE LAW OF ACTION AND REACTION

Actioni contrariam semper et aequalem esse reactionem: sive corporum duorum actiones in se mutuo semper esse aequales et in partes contrarias dirigi.

To every action there is always opposed an equal reaction; or, the mutual actions of two bodies upon each other are always equal, and directed to contrary parts.

Newton's examples: if you press a stone with your finger, the finger is also pressed by the stone. If a horse draws a stone tied to a rope, the horse is equally drawn back towards the stone. The forces always act on *different* bodies – which is why they do not cancel. Equal forces for equal times give equal and opposite changes of momentum, so the **total momentum is conserved** and the common centre of gravity does not move (Corollaries III and IV). In the animation the common centre of gravity is marked with a gold ring: watch it stay still, whatever the masses.

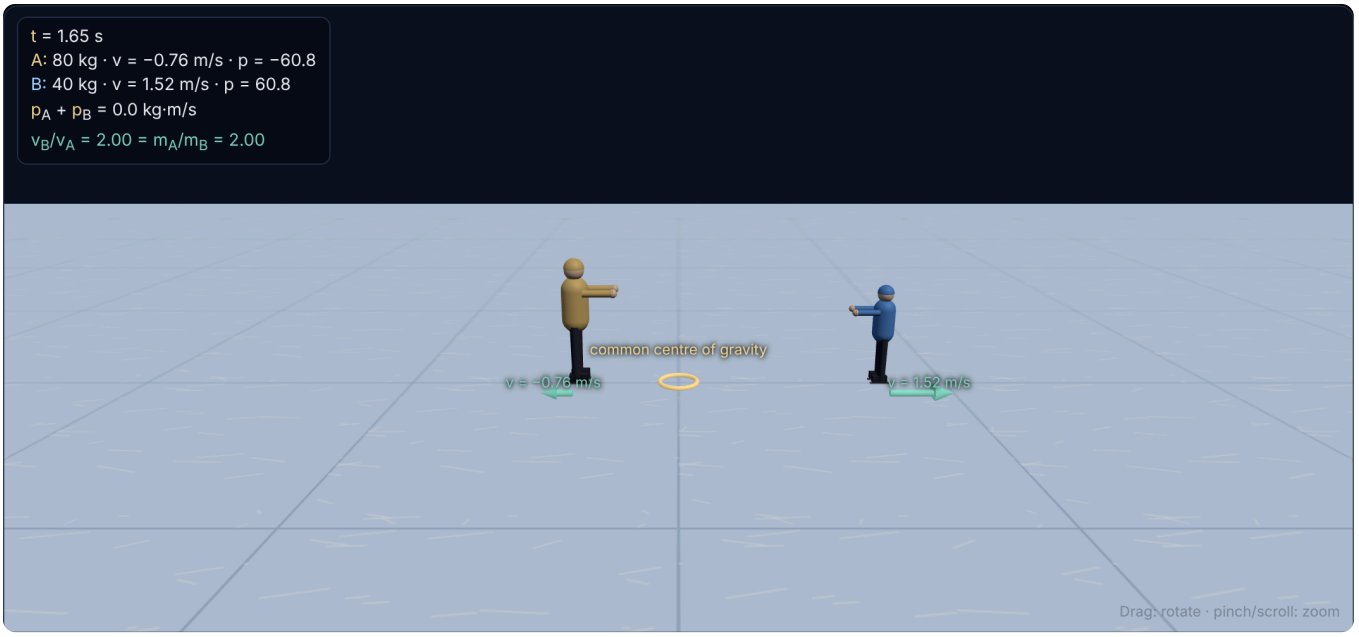


Figure 3. Lex III – the law of action and reaction – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/principia-en.html#lex3

Try it yourself.

1. Make A twice as heavy as B. Are the force arrows equally long? Who gets the greater speed – and how much greater?
2. Why is $p_A + p_B$ always zero? What does that mean for the gold ring?
3. The rocket: it pushes on nothing outside itself. So what does it push on? Follow the centre of gravity as the gas spreads out.

The corollaries – the first consequences

To the laws Newton attaches six corollaries. Three of them have become the backbone of physics:

- **Corollary I – the parallelogram of forces.** Two forces acting together produce the same motion as their diagonal. Forces, in other words, are vectors.
- **Corollaries III and IV – conservation of momentum.** Mutual forces do not change the total momentum, and the common centre of gravity of a system is at rest or in uniform straight-line motion, whatever the parts do to each other.
- **Corollary V – the principle of relativity.** Motions within a space are the same whether that space is at rest or moving uniformly in a straight line – Newton's example is a ship. This is what we now call Galilean invariance: no mechanical measurement can reveal uniform motion.

The laws are followed by a scholium in which Newton shows that they agree with Galileo's laws of falling bodies and projectiles and with the collision experiments of Wren, Wallis and Huygens – he has measured collisions between pendulum bobs himself and corrected for air resistance.

Book I – *De motu corporum*: motion without resistance

The first book is pure mathematics: how do bodies move under centripetal forces? It begins with a section on limits ("first and last ratios"), and then comes the famous **Proposition 1, Theorem 1**:

> *The areas which revolving bodies describe by radii drawn to an immovable centre of force lie in immovable planes and are proportional to the times.*

The proof is one of the most beautiful in physics, and it is exactly Newton's impulse version of the second law. Let the body move uniformly for a time Δt (first law). Then give it an impulse directed towards the centre (second law), and let it again move uniformly. Each of the triangles formed from the centre to two successive points has the same area: the triangle after the impulse has the same base and height as the triangle the body *would* have described without it – because the impulse is parallel to the radius. Now let Δt tend to zero: the polygon becomes a smooth curve, and the impulses a continuous force. Kepler's second law thus holds for *any* central force – it is really the conservation of angular momentum.

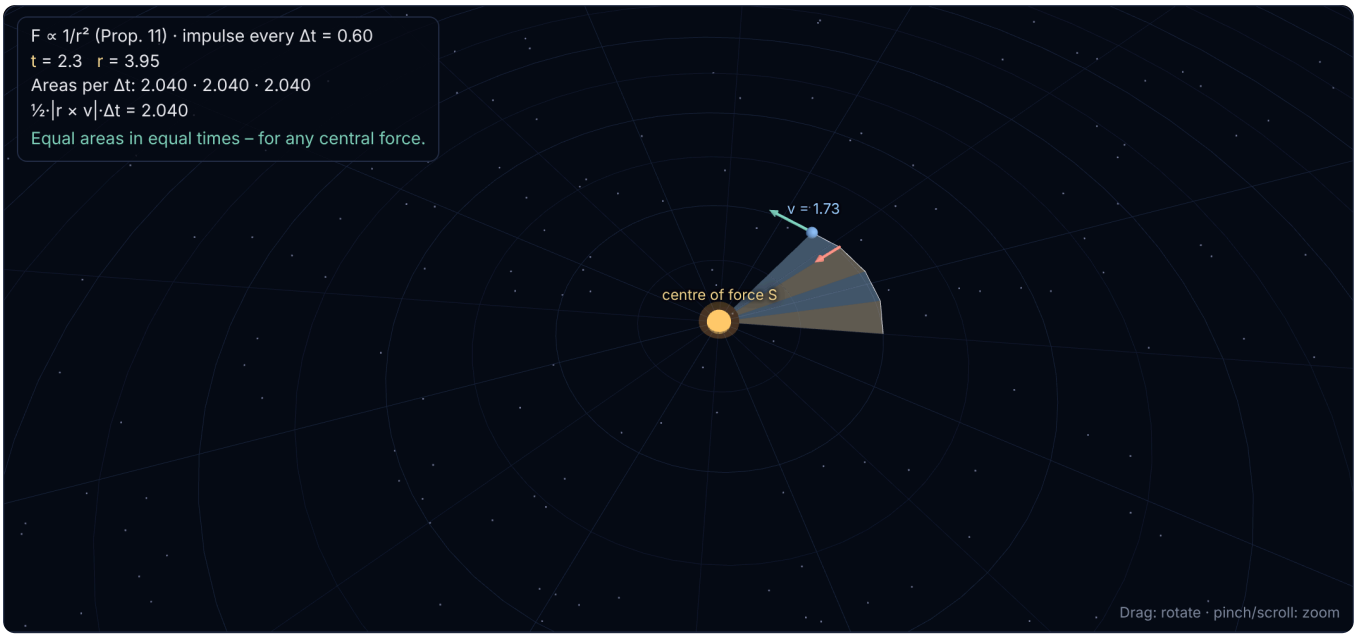


Figure 4. Book I – De motu corporum: motion without resistance – snapshot of the interactive simulation.

Interactive version: https://mhgfysik.dk/principia-en.html#bog1

Try it yourself. The triangles, alternately gold and blue, each cover a time Δt; their areas are shown at the top.

1. Are the areas equal, even when the orbit is very elongated? Are they still equal when you change the force law?
2. Make Δt large in impulse mode. The orbit becomes angular and swings round – yet the areas are still exactly equal. Why?
3. Compare $1/r^2$ and r: in both cases the orbit is an ellipse – but where in the ellipse is the centre of force? (Prop. 10 versus Prop. 11.)

Prop. 11: the ellipse demands $1/r^2$

If a body moves in an ellipse with the centre of force at one focus, the force must fall off as $1/r^2$. This is Halley's question answered – in reverse: here Newton shows that Kepler's first law *requires* an inverse-square law. Prop. 10 shows that a force proportional to distance also gives an ellipse, but with the centre of force at the centre of the ellipse. Prop. 13 extends this to parabolas and hyperbolas, and Prop. 15 gives Kepler's third law: $T^2 \propto a^3$.

Prop. 71: the shell theorem

To compare the Moon's fall with the apple's, Newton had to know that a sphere attracts as if all its mass were at the centre. He proves this in Prop. 71 (and Props. 74–76 for solid spheres): a uniform spherical shell attracts an outside body with $1/r^2$ towards its centre. Inside the shell the force is zero (Prop. 70). This is not self-evident – and it holds only for $1/r^2$ and r.

The rest of Book I deals with, among other things, rotating orbits (Props. 44–45: apsidal motion, with which Newton can test the exponent of the force law from the Moon's perigee), two bodies about their common centre of gravity (Props. 57–63) and the three-body problem (Prop. 66, with 22 corollaries on the Moon's inequalities). Newton even notes at the end that the refraction of light can be understood in terms of particles and forces.

Book II – motion in resisting media

Book II is the most technical and the least read. Newton studies bodies slowed by resistance proportional to velocity, to the square of velocity and to combinations of the two. He measures air resistance by dropping balls inside St Paul's Cathedral and swinging pendulums in air, water and mercury. The book also contains hydrostatics, the first theory of viscosity (Section 9, the hypothesis behind today's "Newtonian fluid") and an attempt to calculate the speed of sound from the elasticity and density of the air (Props. 48–50).

The speed of sound – the famous error

Newton found that the speed of sound is the square root of pressure divided by density, $c = \sqrt{(p/\rho)}$. That gives about 290 m/s, whereas measurements gave about 340 m/s. In the second edition he patched the gap with the "thickness of the air particles" and vapours in the air – a rare example of Newton adjusting his theory to fit the answer. The correct explanation came with Laplace in 1816: the sound wave is so fast that heat has no time to even out, so the process is adiabatic rather than isothermal. That gives $c = \sqrt{(\gamma p/\rho)}$ with $\gamma = 1.40$, and then it fits.

The vortices die

The real target of Book II is Descartes' vortex theory, in which the planets are carried round by a swirling aether. In Section 9 Newton calculates rotating fluids and shows that a vortex cannot give Kepler's third law – the periods would go as r^2 rather than $r^{3/2}$ – and cannot explain why the planets move fastest at perihelion. The book ends with the famous verdict that the vortex hypothesis conflicts with the astronomical phenomena and serves rather to confuse than to explain the heavenly motions.

Book III – *De mundi systemate*: The System of the World

In Book III Newton applies his mathematical theorems to nature. He begins with four **rules of reasoning in philosophy** (*Regulae philosophandi*); the fourth was added only in the third edition:

1. We are to admit no more causes of natural things than such as are both true and sufficient to explain their appearances.
2. Therefore to the same natural effects we must, as far as possible, assign the same causes – respiration in a man and in a beast, the descent of stones in Europe and in America, the light of a kitchen fire and of the Sun.
3. Qualities found in all bodies within the reach of our experiments are to be esteemed the universal qualities of all bodies whatsoever.
4. Propositions inferred by induction from phenomena are to be looked upon as true until other phenomena show them to be more exact or liable to exceptions.

Then come six "phenomena" (among them that the moons of Jupiter and Saturn and the planets obey Kepler's third law, and that the planets obey the law of areas about the Sun) – and then the propositions. The central idea is that the force that keeps the Moon in its orbit *is* the same as gravity on Earth.

The Moon test (Prop. 4)

The Moon is about 60 Earth radii from the Earth's centre, while the apple is 1 Earth radius from it. If gravity falls off as $1/r^2$, the Moon's acceleration towards the Earth must be $60^2 = 3600$ times smaller than g. Newton calculates the Moon's fall from its period and distance and finds that the Moon falls about 15 Paris feet (4.9 m) towards the Earth in one minute – exactly what the apple falls in one second. With modern numbers:

Quantity	Value
Distance of the Moon, $r = 60.3\ R_E$	3.844×10^8 m
Sidereal period, T	27.32 days = 2.361×10^6 s
The Moon's centripetal acceleration, $4\pi^2 r/T^2$	2.72×10^{-3} m/s²
Surface gravity diluted, $g/60.3^2$	2.70×10^{-3} m/s²
The Moon's fall in 1 minute, $\tfrac{1}{2} \cdot a \cdot (60\ \text{s})^2$	4.9 m – like the apple in 1 s

From that one law, Book III goes on to explain:

Universal gravitation (Prop. 7)

Gravity acts between all bodies and is proportional to the quantity of matter in them. By the third law the Earth must also pull on the Sun and the Moon. Newton determines the ratio of the masses of the Sun, Jupiter, Saturn and the Earth from the orbits of their moons – and concludes that it is the system's centre of gravity, not the Sun, that is at rest. The Sun itself wobbles slightly. He did not know the gravitational constant *G*; it was first measured by Cavendish in 1798.

The shape of the Earth and precession (Props. 18–20, 39)

A rotating fluid Earth must be flattened at the poles. Newton calculates the flattening as 1/230 (today 1/298) and predicts that a pendulum clock runs slower at the equator – which Jean Richer had already observed in Cayenne in 1672. The equatorial bulge of the flattened Earth is pulled by the Sun and the Moon, so the Earth's axis traces out a cone: the precession of the equinoxes, discovered by Hipparchus.

The tides (Props. 24, 36–37)

The tide is the difference between the Moon's (and the Sun's) pull on the near and the far side of the Earth. This explains two high tides a day, spring and neap tides, and why the Moon matters more than the Sun, even though the Sun pulls far harder on the Earth as a whole. See Tides in 3D.

Lunar theory and the comets

Props. 22–35 treat the Sun's disturbance of the Moon's orbit: the regression of the nodes, the variation and the changing eccentricity of the orbit (see The Moon's orbit in 3D). The final part shows that comets are celestial bodies in very elongated orbits. Newton fits a parabola to the great comet of 1680 – one of the main proofs that the heavens obey the same laws as the Earth.

The famous drawing of a cannon on a mountain firing its ball faster and faster until it goes into orbit is not in the *Principia* itself. It comes from Newton's popular version, *De mundi systemate* (published after his death, in 1728). You can try it here: Newton's cannon in 3D.

The General Scholium – *Hypotheses non fingo*

In the second edition (1713) Newton ended the book with a General Scholium. He repeats his criticism of the vortices and writes that this most beautiful system of the Sun, planets and comets could only

proceed from the counsel and dominion of an intelligent and powerful Being. A dense passage follows on God as Lord over all – not the soul of the world, but its ruler – eternal, infinite and omnipresent.

Most quoted is the ending on gravity. Newton has shown *how* gravity acts, but cannot discover its *cause* from the phenomena:

> *Hypotheses non fingo – I frame no hypotheses.*

It is enough, he writes, that gravity really exists, acts according to the laws he has set out, and accounts for the motions of the heavenly bodies and of the sea. Leibniz and Huygens criticised a force acting at a distance through empty space as an "occult quality". The open question received a new answer only with Einstein's general theory of relativity in 1915, in which gravity is the curvature of spacetime.

What the *Principia* changed

The book was hard to read; according to a well-known anecdote a student remarked as Newton passed by: "There goes the man that writ a book that neither he nor anybody else understands." But on the Continent Varignon, Johann Bernoulli, Euler, Lagrange and Laplace translated the geometrical proofs into differential equations, and during the eighteenth century Newton's mechanics became the analytical mechanics we teach today.

- **1735–1744:** French expeditions to Lapland (with Anders Celsius) and Peru measure a degree of meridian and confirm that the Earth is flattened at the poles – against the Cartesian prediction.
- **1758–59:** Using Newton's theory, Halley had predicted that the comet of 1682 would return. It was recovered on Christmas Day 1758 and reached perihelion in March 1759. Halley had been dead for 16 years.
- **1846:** Le Verrier and Adams predict a new planet from the deviations of Uranus. Galle finds Neptune less than 1° from the calculated position.
- **1905 and 1915:** Einstein shows where Newton's mechanics ceases to hold – at speeds close to that of light and in strong gravitational fields (for example the perihelion advance of Mercury). But in the limit his theories become Newton's, and it is still Newton's laws that steer spacecraft to Pluto.

Sources and further reading

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